School of Computing & Information Sciences
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Engineering & Computing
School of Computing & Information Sciences

Doe - Menstral Cycle Tracker

Students: Monica Barbosa, Gabriel Garcia, Kristian Correa, Marcelo Hernai
, *Florida Internaion University*
Instructor: *Dr. Masoud Sadjadi*, Florida International University

Problem

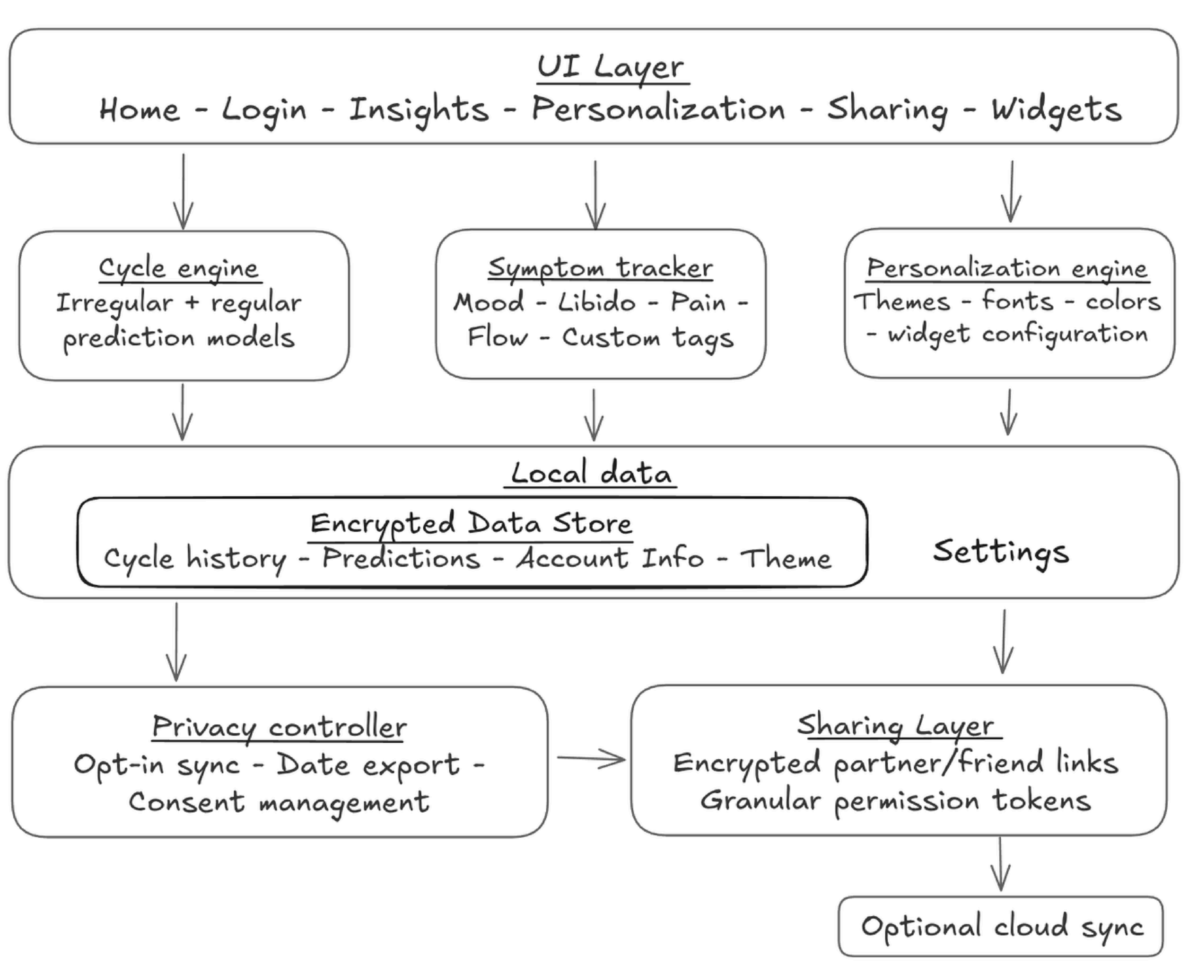
Users rely on menstrual cycle tracking apps to monitor reproductive health, fertitility, and symptoms. However, these apps often require users to store highly sensitive health data on third-party cloud servers. This creates trust and safety concerns, especially in environments where reproductive health data could be legally or socially sensitive (Especially after the overturning of Roe v. Wade in 2022)

Because of this, many users:
- Feel unsafe using cycle tracking apps
- Reduce or stop tracking their cycles entirely
- Lose access to useful health insights as a result.

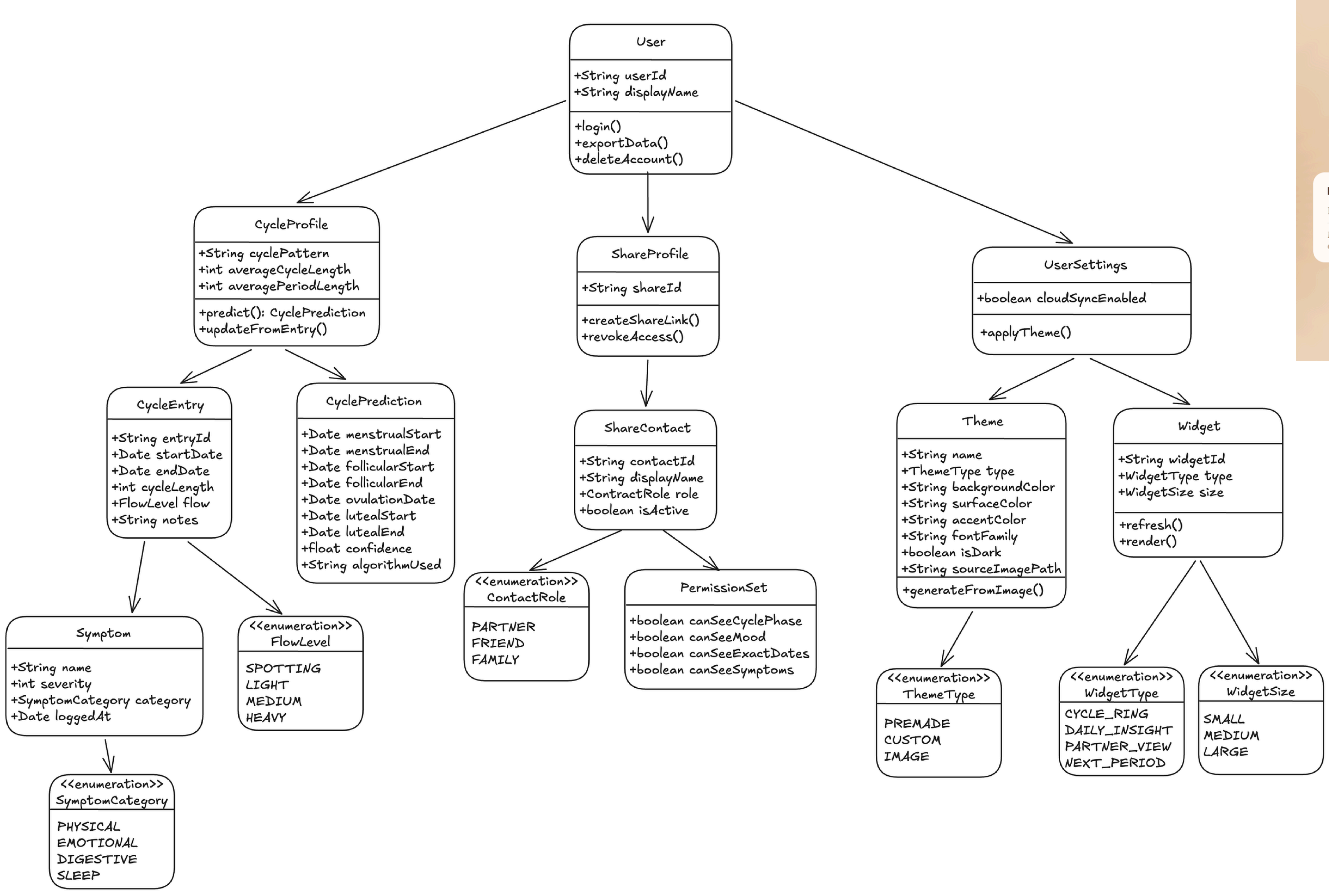
In addition, existing apps fail to meet modern user expectations for:
- Data ownership (users cannot fully control where their data goes)
- Customization (limited personalization and rigist UI designs)
- Inclusive tracking (poor handling of irregular cycles and unique patterns)

As a result, there is still a need for a system that is:
- Fully local-first (data stored on-device by default)
- Customizable beyond UI themes
- Aaptive for both regular and irregular cycle predictions
- Designed with user-controlled, fine-grained data sharing

System Design



Object Design



Current System

Existing applications include:
- Popular menstrual tracking applications such as Flo Period Tracker, Clue Period & Cycle Tracker, Drip, and Stardust Period Tracker provide digital tools for period tracking, symptom logging, and fertility predictions.

System Overview:
- These applications typically operate using cloud-based architectures, where user data is stored on external servers accessed through user accounts
- Prediction models are based on historical cycle data and statistical averaging, often optimized for regular cycle patterns
- Features such as reminders, insights, and sharing are delivered through centralized, company-controlled platforms

Key Limitations:
- Privacy dependency: Sensitive reproductive health data is stored and managed by third-party servers, requiring full user trust in company policies
- Limited data ownership: Users have restricted control over how data is stored, accessed, or potentially shared
- Low customization: Applications offer mostly superficial personalization (themes, colors, UI skins)
- Prediction gaps: Irregular cycles are less accurately modeled compared to regular cycles
- Restricted sharing control: Partner or friend access is often limited and not fully granular

Implementation

- The system is implemented as a mobile application using a modular architecture separating UI, data storage, and prediction logic.
- A local database is used to store all user data on-device by default, with optional cloud synchronization for account-enabled users.
- The prediction engine processes historical cycle data to estimate future cycles for both regular and irregular patterns.
- The user interface is built to support customization features including themes, layouts, and widget integration.

Requirements

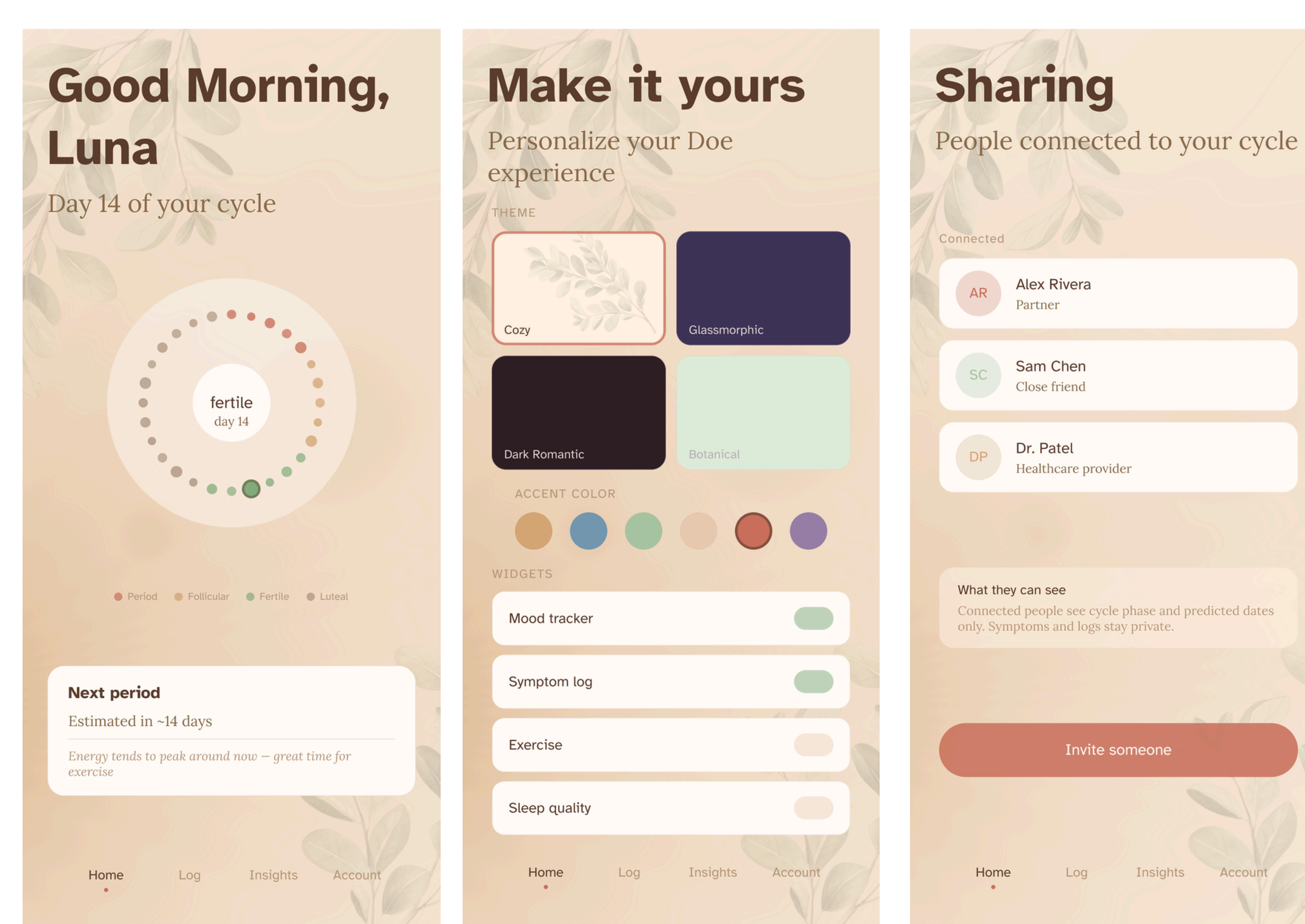
Functional:
- The system shall allow users to record menstrual cycle data and generate predictions for regular and irregular cycles.
- The system shall store all data locally by default and support optional cloud synchronization through user accounts.
- The system shall allow controlled sharing of selected data with partners or trusted individuals.
- The system shall provide customization options and interactive widgets for cycle tracking and reminders.

Non-Functional:
- The system shall follow a privacy-first design in which it prioritizes privacy by storing data locally unless explicitly shared or synced.
- The system shall ensure security through encrypted data storage.
- The system shall provide fast and reliable performance for data processing and predictions.
- The system shall maintain a simple, intuitive, and accessible user interface.
- The system shall be scalable to support future feature enhancements.

Verification

- Unit testing is used to validate cycle prediction logic for both regular and irregular inputs.
- Integration testing ensures correct interaction between the user interface, local database, and prediction engine.
- System testing verifies that data is stored correctly and remains persistent across sessions.
- Security testing confirms that user data remains local unless explicit permission is granted for synchronization.
- User testing is conducted to evaluate usability, customization features, and overall experience.

Screenshots



Summary

This project presents a privacy-focused menstrual cycle tracking system designed to give users full control over their personal health data. Unlike traditional cloud-based applications, the system prioritizes local data storage by default while still supporting optional synchronization when desired. It also introduces enhanced customization features and flexible sharing options for partners or trusted individuals. In addition, the system supports both regular and irregular cycle prediction to improve inclusivity and usability. Overall, the application provides a modern, user-centered alternative to existing menstrual tracking solutions by combining privacy, personalization, and functionality.

Acknowledgement

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